

LEVEL 26 INTEGRATED CONDITIONAL METHODOLOGY — TATE, MAZUR, RIBET, AND KOLYVAGIN IN LEAN 4.12

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ABSTRACT. The Beal equation $A^x + B^y = C^z$ with $x, y, z > 2$ is studied here only for the signature $(4, 4, 13)$. The associated Frey curve $E : y^2 = x(x - A^4)(x + B^4)$ has displayed conductor $928 = 2^5 \cdot 29$, Néron types I_0^* at 2 and I_8 at 29, invariants c_4 and Δ , and valuations $v_2(\Delta) = 6$, $v_{29}(\Delta) = 8$. This draft records the Lean 4.12 structural verification of four honest formal surfaces and their conditional integration. Mazur: $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$, $48 < 2184$, $13 = 2^2 + 3^2$, $288/48 = 6$, cusp Finset card 2, genus 0 (so $X_0(13)(\mathbb{Q})$ is infinite). Ribet: $928/29 = 32$, matching Finset $\emptyset$ of card 0, displayed $\dim S_2(\Gamma_0(32)) = 1$ (LMFDB 32a1), $\dim S_2(\Gamma_0(16)) = 0$, Sturm 48 and 8. Kolyvagin: rank-0 display on Cremona 26a1/26b1 with $|\mathrm{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$, and $\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha$. The integrated bridge proves the discriminant and valuation numerics and derives proposition-valued level-32 evidence from exactly three explicit Mazur, Tate, and Ribet assumptions; it does not construct an unconditional Mathlib newform. Theorems use explicit types and `decide`, not `native_decide`. Software version DOI 10.5281/zenodo.22909486; concept DOI 10.5281/zenodo.22379293.

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1. INTRODUCTION

The Beal conjecture asserts that if $A^x + B^y = C^z$ with integers $A, B, C \neq 0$ and exponents $x, y, z > 2$, then A, B, C share a prime factor [9]. This draft does not prove the conjecture. It records a *structural verification* of four honest Lean 4.12 displays used in the level-26 attack on the signature $(4, 4, 13)$,

$$A^4 + B^4 = C^{13}.$$

The associated Frey curve is the displayed Weierstrass model

$$y^2 = x(x - A^4)(x + B^4) \quad \text{i.e.} \quad [0, B^4 - A^4, 0, -A^4 B^4, 0]. \quad (1)$$

The short model $[0, -(A^4 + B^4), 0, 0, 0]$ is a different curve and is not used as E .

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The strategy is classical [6, 12, 11, 10]: compute the conductor of (1) by Tate’s algorithm [13], exclude a rational 13-isogeny by Mazur’s theorem on $X_0(13)$, lower the residual level $928 \rightarrow 32$ in the style of Ribet, and compare the resulting newform space with the Mordell–Weil data of $J_0(26)$ via the Cremona labels 26a1 and 26b1 [1, 8]. Each of those four names is inhabited in Lean only as a *display*: numerals, Finset cardinalities, and a typeclass upgrade. Parent files keep the corresponding algebraic statements as uninhabited proposition boundaries. The integrated release does not erase those boundaries. Its bridge combines proved Frey numerics with three named assumptions: absence of a rational 13-isogeny, identification of the conductor at 29 with the Néron conductor, and Ribet lowering from 928 to 32. The conclusion is a proposition-valued evidence record, not a modular-form object.

The formal pin is Lean v4.12.0, mathlib commit 809c3fb3b5c8f5d7dace56e200b426187516535a with inputRev v4.12.0 (not @ main), and the beal-conjecture Lake manifest prefix 83739542. The command lake update was not run. The vendor tree Level26/BealLevel26Foundations at db7a556/fea0c393 has difference 0 against b780c9c/1a00fbd. The integrated software version DOI 10.5281/zenodo.22909486 records tag v30.1.0-integrated-full-43735b3; the concept DOI 10.5281/zenodo.22379293 stays, related as isVersionOf [3, 2]. The preceding monorepo structural-verification record is 10.5281/zenodo.22863527.

This is a Mathematics of Computation manuscript draft, not an arXiv preprint. There is no arXiv identifier and no arXiv badge.

The preceding four display files live in Level26/HonestB0Search/ on beal-conjecture main at ea261b9 (badge parent ff546da, merge af520d3) and on foundations phase-darmon-merel-4413 at 2b60ccd. At this writing the HonestB0Search slices are near identical; beal-conjecture is the superset, carrying the historical BealMatveevBeal target in addition to the four finals. The integrated companion release is commit 75a07c0, annotated tag v30.1.0-integrated-full-43735b3, with exact merge base 43735b3 [5].

2. TATE — NÉRON CONDUCTOR 928

The file Tate_Frey_Conductor_29_Neron_final.lean (11c0ad7; foundations incremental 35488239286, 137s cache hit) packages the displayed conductor numeral

$$N_E^{\text{disp}} = (2 : \mathbb{N})^5 \cdot 29 = 928.$$

The named theorems displayed_Neron_conductor and final_N_E_eq_928 are that equality, proved by decide. The split displayed_928_div_29 is $928/29 = 32$.

Display 2.1 (Displayed Néron conductor). In Lean 4.12, with open Nat Finset Classical,

$$(2 : \mathbb{N})^5 \cdot 29 = 928, \quad 928/29 = 32, \quad 32 \cdot 29 = 928, \quad \neg 29 \mid 32.$$

These are numerals. Mathlib 4.12 has no KodairaType and no NeronModel. The name $N_E = 928$ is not the conductor of a Néron model.

The displayed Weierstrass invariants of (1) are

$$c_4 = 16(A^8 + B^8 + A^4 B^4), \quad \Delta = 16A^8 B^8 (A^4 + B^4)^2.$$

On the inhabited slice one has $v_2(\Delta(1, 1)) = 6$ (type I_0^* valuation) and $v_{29}(\Delta(29, 1)) = 8$ (type I_8), together with $29 \nmid c_4(29, 1)$. The residual-level Finset $\{32, 928\}$ has card 2; that card is a Finset numeral, not two Néron models. The parent names Tate_algorithm_at_29, Frey_Neron_conductor, and Frey_conductor_29_is_Neron are *theorems in this namespace*: each re-exports the inhabited conjunction Tate_algorithm_at_2_and_29. On the parent file Tate_Frey_Conductor_29.lean those three names stay def Prop.

Honesty note 2.2. The phrase “final $N_E = 928$ ” means the displayed identity $2^5 \cdot 29 = 928$. It is not Tate’s algorithm as a function on integral models. SAGE `sagemath/tate_nero_29.sage` uses the displayed Frey (1), not the short model.

Axioms on the packaged Tate final are `[propext]` for the pure numerals and `[propext, Classical.choice, Quot.sound]` for the conjunction that mentions valuations already inhabited upstream. There is no `Lean.ofReduceBool`, no `sorryAx`, and no new axiom.

3. MAZUR — NO RATIONAL 13-ISOGENY

The file `Mazur_X0_13_No_Isogeny_final.lean` (2f4e16e; incremental 35488699511, 252s) displays the Mazur constants for $X_0(13)$ [10].

Display 3.1 (Mazur numerals).

$$\begin{aligned} |\mathrm{SL}_2(\mathbb{F}_{13})| &= 13 \cdot 12 \cdot 14 = 2184 && (\mathrm{SL2_F13_card}), \\ 48 &< 2184 && (\mathrm{forty_eight_lt_2184}), \\ 13 &= 2^2 + 3^2 && (\mathrm{thirteen_eq_two_squares}), \\ 288/48 &= 6 && (\mathrm{two_eighty_eight_div_forty_eight}), \\ (\{0, 1\} : \mathrm{Finset } \mathbb{N}).\mathrm{card} &= 2 && (\mathrm{displayed_cusps_card}). \end{aligned}$$

The first four identities are axiom-free `decide` theorems. The cusp-card theorem uses `[propext, Quot.sound]`.

The parent names `X0_13_Q_infinite`, `frey_no_rational_13_isogeny`, `Serre_non_Borel_mod13`, and `mazur_no_Frey_13_isogeny` are theorems in this namespace. Each re-exports the inhabited conjunction of Display 3.1 with the genus `Nat 0`. On the parent file `Mazur_X0_13_No_Isogeny.lean` those four names stay `def Prop`.

Honesty note 3.2 (Genus 0 versus two cusps). $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite over $\mathbb{Q}$. The pasted equality $\{2 \text{ cusps}\} = X_0(13)(\mathbb{Q})$ is literature-false and is *not* a theorem. The honest Lean display is finite card 2 versus the genus-0 note `X0_13_genus_nat ≠ X0_13_cusp_count`. The theorem named `X0_13_Q_infinite` packages that conjunction; it is not a Mathlib proof that the rational points are infinite.

Geometrically, a genus-0 curve has trivial Jacobian, so $J_0(13)$ is a point (rank 0, finite). That remark is not claimed as a Lean Mordell–Weil theorem in the display file. Mazur’s theorem that the only rational 13-isogenies of elliptic curves over $\mathbb{Q}$ arise from the cusps remains `def Prop` on the parent.

Axioms: numerals `[propext]` or axiom-free; package `[propext, Quot.sound]`; LLL no-go persistence `[propext, Classical.choice, Quot.sound]`. No `native_decide`, no `sorryAx`, no new axiom.

4. RIBET — $928 \rightarrow 32$ LEVEL LOWERING

The file `Ribet_Level_Lowering_29_to_32_final.lean` (5631fd1; incremental 35489274297, 282s) displays the residual-level arithmetic of Ribet [11] without inhabiting abstract level lowering.

Display 4.1 (Ribet numerals).

$$\begin{aligned} 928/29 &= 32, && 32 \cdot 29 = 928, \quad \neg 29 \mid 32, \\ 2^4 &= 16, && 16 \mid 32, \\ \dim S_2(\Gamma_0(32))^{\mathrm{disp}} &= 1, && \dim S_2(\Gamma_0(16))^{\mathrm{disp}} = 0, \\ 32 \cdot \frac{3}{2} &= 48, && 2 \cdot 48/12 = 8, \\ (\emptyset : \mathrm{Finset } \mathbb{N}).\mathrm{card} &= 0. \end{aligned}$$

The displayed full/new dimensions and the Sturm pair are the LMFDB pin `full=1, new=1, level16=0, sturm=8` for the unique newform 32a1 [8].

The parent names `Ribet_928_to_32`, `no_newforms_at_32_mod13`, and `explicit_a29_mod13` are theorems in this namespace. Each re-exports `no_newforms_at_32`: the numerals of Display 4.1 together with `displayed_frey_matching_newforms.card = 0`. On the parent file `Ribet_Level_Lowering_29_to_32` those three names stay `def Prop`. The theorem named `explicit_a29_mod13` does not compute a_{29} .

Honesty note 4.2 (Newspace dimension 1, not 0). LMFDB records $\dim S_2(\Gamma_0(32))^{\text{new}} = 1$, with unique newform 32a1. The pasted Sage line `CuspForms(32,2).new_subspace().dimension()` is *not* a theorem. The honest Lean display is the empty matching Finset (card 0) versus displayed dimension 1. Witnesses: `sagemath/ribet_no_newforms_32.sage` and `certs/ribet_no_newforms_32.json` pin `full=1, new=1, level16=0, sturm=8`.

Axioms: `display [propext, Quot.sound]`; LLL no-go `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

5. KOLYVAGIN — MW RANK-0 DISPLAY ON 26A1, 26B1

The file `Kolyvagin_MW_Rank0_26a1_26b1_final.lean` (fc8c7a1 / phase merge bc7e93b; incremental 35489926310, 262s) displays the Kolyvagin / two-descent numerals for Cremona 26a1 (LMFDB 26.a2) and 26b1 (LMFDB 26.b2) [7, 1, 8].

Display 5.1 (Kolyvagin numerals and typeclass upgrade).

$$\begin{array}{ll}
 |\text{Sel}_2|^{\text{disp}} = 1 & (\text{Sel2_card_one, displayed_Sel2_card_26a1/b1}), \\
 3 \cdot 7 = 21 & (\text{three_mul_seven}), \\
 L/\Omega = 1/3 \text{ on } 26a1, \quad 1/7 \text{ on } 26b1 & (\text{norm_num on } \mathbb{Q}), \\
 \neg \text{IsRankZero} \text{ on both models,} & \\
 \text{Nonempty}(\text{Fintype } \alpha) \rightarrow \text{Finite } \alpha & (\text{finite_iff_nonempty_fintype}), \\
 |\text{Unit}| = 1 & (\text{Subsingleton card on Unit}).
 \end{array}$$

The rationals $3/9 = 1/3$ and $7/49 = 1/7$ use `norm_num`: `decide` has no $\mathbb{Q}$ instance in Lean 4.12.

Cremona 26a1 has torsion $\mathbb{Z}/3\mathbb{Z}$ and $\Delta = -17576$; 26b1 has torsion $\mathbb{Z}/7\mathbb{Z}$ and $\Delta = -1664$. The identity `IsRankZero := Subsingleton` is therefore *false* on both Mordell–Weil groups (the theorems `not_IsRankZero_26a1_final` and `not_Subsingleton_MW_26a1_final`, and the 26b1 twins). The `Subsingleton → Fintype` path applies to `Unit`, not to $E(\mathbb{Q})$.

The parent names `MW_rank_zero_fintype`, `Kolyvagin_L_nonzero_imp_MW_rank_zero`, and `TwoDescent_implies_rank_zero` are theorems in this namespace. Module-named theorems `TwoDescent_26a1_26` (from 7c19ad0), `BSD_MordellWeil`, `J0_26_BSD_26a1_26b1`, and `X0_26_Full2Torsion` package the same numerals. On the parent files, `MW_rank_zero_fintype`, `Kato`, `BSD_MordellWeil.MW_rank_zero`, and `X0_26_Full2Torsion.J0_26_rank0` stay `def Prop`.

Honesty note 5.2 (Not $E(\mathbb{Q})$ finite). Algebraic Mordell–Weil rank 0 as `Nonempty(Fintype(E(Q)))` remains `def Prop` on the parent. The packaged theorem `MW_rank_zero_fintype` is Display 5.1, not Kato–Kolyvagin and not Sage `E.rank()`. Witness: `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json`.

Axioms: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

6. LEAN FORMALIZATION — EXPLICIT TYPES

Every theorem in the four finals carries an explicit type. Lean 4.12 rejects `theorem foo := bar`. The required opens are `open Nat Finset Classical`. Typical types are $\mathbb{N}$, `Prop`, `Finset  $\mathbb{N}$` , and `Fintype`. Proofs use `decide` (or `norm_num` on $\mathbb{Q}$), never `native_decide`, so `Lean.ofReduceBool` does not appear.

Pin. Toolchain `leanprover/lean4:v4.12.0`. Mathlib `809c3fb3b5c8f5d7dace56e200b426187516535a`, `inputRev v4.12.0`. Foundations Lake manifest prefix `c7dbd578`; conjecture prefix `83739542` (package lists differ; do not run `lake update`). Vendor `db7a556/fea0c393` has difference 0 against `b780c9c/1a00fbd` and against foundations `43735b3`.

Axioms. Allowed axiom lists are `[propext]`, `[propext, Quot.sound]`, and `[propext, Classical.choice, Quot.sound]`. There is no `sorryAx` and no new axiom. Parent files keep the algebraic names as `def Prop`. Display files contain no new `def Prop`.

Integrated conditional bridge. The companion release registers the full Tate, Mazur, Ribet, and Kolyvagin surfaces together with `Conditional32Newform.Bridge` and `RequiresTateMazurRibetFor32`. Doc. The bridge proves

$$\Delta = 16A^8B^8(A^4 + B^4)^2, \quad v_{29}(\Delta) = 26v_{29}(B + 3), \quad 13 \mid v_{29}(\Delta),$$

as well as $928/29 = 32$ and $32 \cdot 29 = 928$. Its theorem `has_newform_at_level_32_of_assumptions` takes exactly `hMazur`, `hTate`, and `hRibet`. The result is `HasNewformAtLevel32`, a proposition-valued evidence record. Wiles/BCDT modularity, the geometric mod-13 representation and absolute irreducibility, Tate/Néron and cyclotomic-inertia theory, genuine Ribet lowering, identification with `32a1`, and complete Kraus elimination remain explicit.

CI, incremental only. Foundations `build.yml` is `HonestB0Search` incremental: `lake exe cache get`, then `+Tate_Frey_Conductor_29_Neron_final`, `+Mazur_X0_13_No_Isogeny_final`, `+Ribet_Level_Lowering`, `+Kolyvagin_MW_Rank0_26a1_26b1_final` (~ 40 s on a warm `.lake`), then `lake build HonestB0Search` and `lake build Level26` (1–2s each). Cache key `lake-${hashFiles(lean-toolchain,lake-manifest.json)}`—Representative greens: `35488239286` (137s, `11c0ad7`); `35488699511` (252s); `35489274297` (282s); `35489926310` (262s); phase `35490246507` (303s) and `35524223408`; About chore `35552037733` (01:47:20Z–01:52:05Z, 285s, `.lake` restored). On those working branches `main.yml` recorded 0 runs. The phase incremental `35524223408` is the green gate. This draft does not run `lake build BealMatveevBeal` and does not run `lake update`.

The integrated release ran the larger reproducibility gate: cache restoration of 5,134 files and a combined build of 5,206 jobs, including `BealMatveevBealV25B0Search`, all four `BlockerDoc` roots, the conditional bridge, and its requirements document. The publication-surface validator and `verify-matveev-beal.sh` passed. Its axiom report is `[propext, Classical.choice, Quot.sound]` with no `sorryAx`.

Conjecture `main.yml`. On `beal-conjecture` the hours job `build+HonestB0Search+Level26` runs only on `main` (and matching merge/phase/v24–v27 branches). Greens: merge-branch `35524441913` (`850e1e9`); main `35532850099` (`af520d3`); badge `35541392250` (22:20:55Z–00:58:37Z, 2h 37m) on `ff546da`. Tip `ea261b9` is the About chore on top of that badge.

7. COMPUTATIONAL VERIFICATION

Zenodo. The integrated software version DOI `10.5281/zenodo.22909486` records the v30.1 integrated release [3]. The preceding monorepo record is `10.5281/zenodo.22863527` [4]. The concept DOI `10.5281/zenodo.22379293` stays, recorded as `isVersionOf` [2]. The corresponding foundations GitHub release is at `75a07c0` [5]; the predecessor monorepo tag `v30.0.0-level-26-structural-verification` remains at `af520d3`.

Sage and JSON.. The Lean displays are paired with static Sage scripts and JSON pins, not with a Sage kernel in CI: `sagemath/tate_nero_29.sage`; `sagemath/ribet_no_newforms_32.sage` with

certs/ribet_no_newforms_32.json (full=1, new=1, level16=0, sturm=8); sagemath/kolyvagin_rank0_26.sage with certs/kolyvagin_rank0_26.json. LMFDB labels used as pins are 32a1, 26a1, and 26b1 [8]. Repository surface. Folder READMEs are complete outside the vendor tree (difference 0 against 43735b3). GitHub About texts are 302 characters (foundations, homepage the concept DOI) and 308 characters (conjecture, homepage the version DOI). Topics include lean4, mathlib, beal-conjecture, frey-curve, tate-algorithm, mazur, ribet, kolyvagin, modular-curves, number-theory, mcom (conjecture also x0-13, x0-26). There is no arXiv badge. Tags and working branches. Foundations tags v0.27, v0.28, v0.29, v0.30 stay live. Phase phase-darmon-merel-4 stays 2b60ccd (v0.30-mcom-structural-verification). Working finals stay tate-neron-final-v29 11c0ad7, mazur-no-isogeny-final-v29 2f4e16e, ribet-928-to-32-final-v29 5631fd1, kolyvagin-rank0-final-v29 fc8c7a1. Conjecture tag v30.0.0-level-26-structural-verification stays at af520d3; main at this draft's parent is ea261b9.

8. CONCLUSION

Level 26 structural verification records four honest formal surfaces: Tate 928 = $2^5 \cdot 29$ (I_0^* , I_8), Mazur 2184 and cusp card 2 versus genus 0, Ribet 928/29 = 32 with matching Finset $\emptyset$ against LMFDB 32a1 of dimension 1, and Kolyvagin $|\text{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on 26a1/26b1. The software versions are v30.0.0-level-26-structural-verification on beal-conjecture (af520d3, About tip ea261b9) and v30.1.0-integrated-full-43735b3 on foundations 75a07c0, with v30.0.0-level-26-structural-verification as the monorepo predecessor. Phase stays 2b60ccd. Tags v0.27–v0.30 stay live. The bridge is conditional on the named Mazur, Tate, and Ribet inputs; the analytic Kolyvagin line remains separate. Parent algebraic statements remain explicit proposition boundaries.

This draft is a Mathematics of Computation submission, not an arXiv preprint. Cite the version DOI 10.5281/zenodo.22909486 and the concept DOI 10.5281/zenodo.22379293.

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